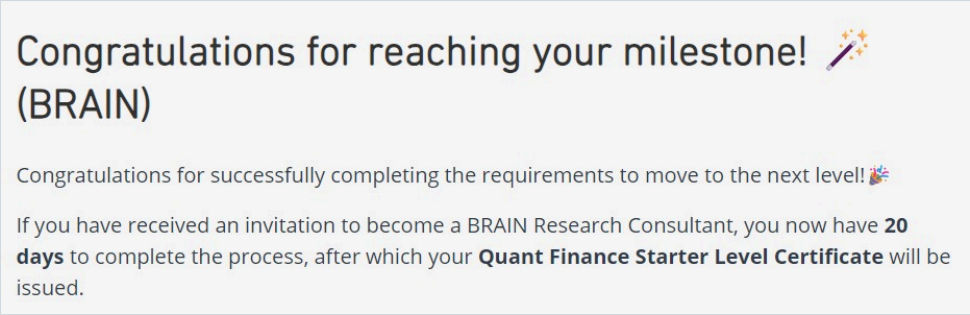
Week 4: Expanding Ideas and Reaching Gold

- Dates: 2026-10-20 → 2026-10-28 (started early, during week 3)
- Scope: BRAIN Tutorial Chapter 6 — Completion: reach 10,000 Challenge points (Gold) ✓
- Simulations: 21, 3 submitted (7 in total)
- Common settings: USA · D1 · TOP3000 · Truncation 0.08 · NaN handling On · Test period 0 (full in-sample period)



Summary

Chapter 6 has one task: keep submitting until the Challenge score reaches 10,000 (Gold). The daily cap is 2,000 points and one alpha earned about 1,900, so it meant one new alpha a day.

The hard part was not performance but **originality**. Variations of an alpha I had already submitted were blocked by the self-correlation check, even when they performed better. Every new submission needed a genuinely new idea. New ideas were usually weak on their own, but blending two or three signals with different characteristics got them over the bar.

Score: 3,876 → 7,690 → 9,587 → **Gold** (confirmed 10/28).

Roadmap

```
Expand the week 3 news alpha (universe / neutralization / signed_power / trade_when)
    up to Sharpe 2.03, but self-correlation 0.94 → blocked
New data: analyst-revision (reversed) + financial health (FFO / debt)
    0.90 + 0.67 → 1.34                        --> SUBMITTED (10/21)
Leverage alpha from week 2, re-checked on the full period
    Sharpe 1.55                               --> SUBMITTED (10/24)
Option IV gap + liability fair value + leverage
    Sharpe 1.90, Fitness 1.81                 --> SUBMITTED (10/25) → Gold
```

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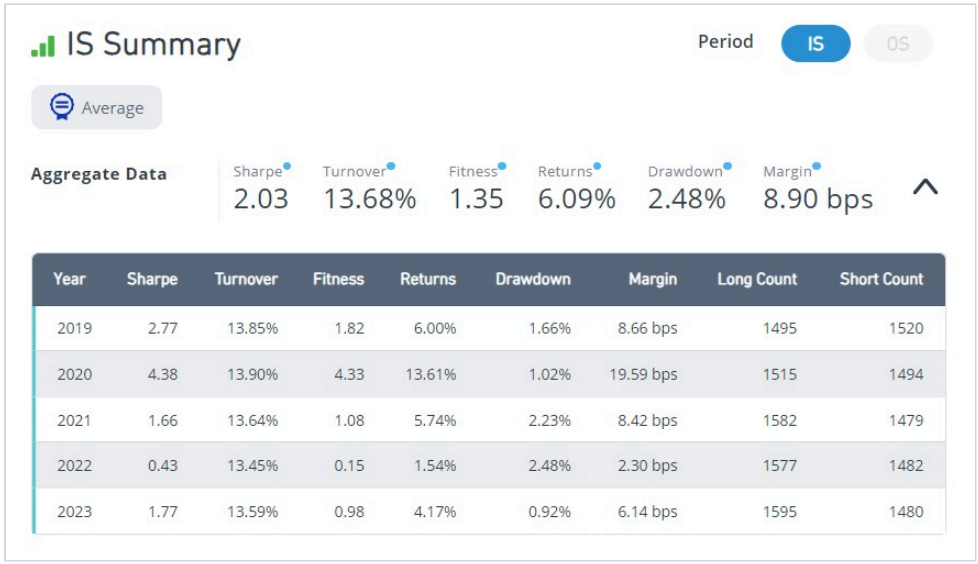
1. Four ways to expand an idea

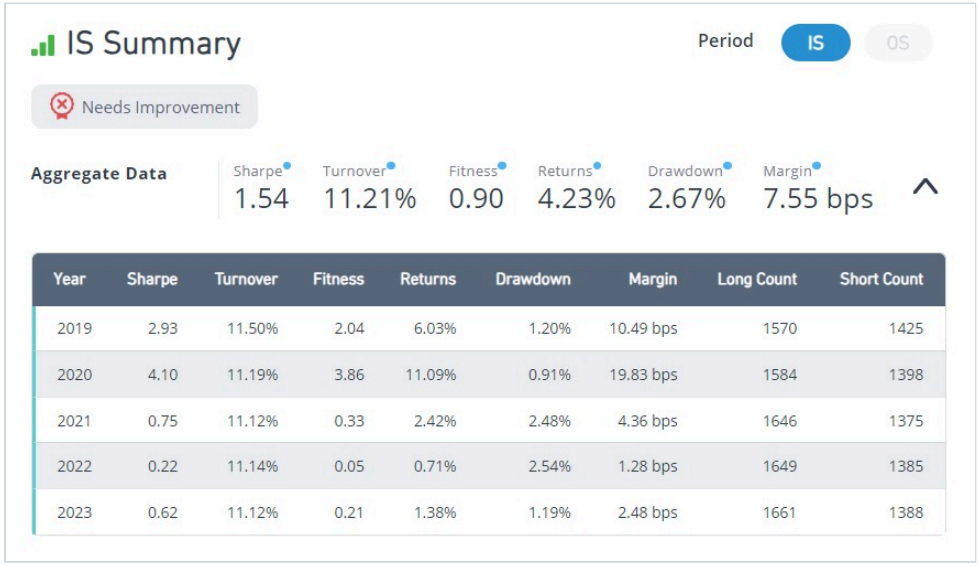
The tutorial lists four ways to get more alphas from one idea: change the universe, change the neutralization, change how positions are distributed (`rank` , `signed_power` , `log`), and combine related alphas (for example `trade_when(A1 > x, A2, A1 <= x)`). It also warns against over-tuning in-sample, and against adding

unrelated alphas together; diversification should be judged by whether one alpha holds up while another is in drawdown.

I applied each method to my week 3 news alpha:

Method	Change	Sharpe	Fitness	Outcome
(baseline)	week 3 submission	1.91	1.25	already submitted
Universe	TOP1000	1.44	0.83	fails — signal lives in smaller stocks
Neutralization	Industry	1.73	1.17	passes thresholds
Position distribution	reversal leg → `signed_power(·, 2)`	**2.03**	**1.35**	best
Synergy	`trade_when(volume > adv20, ·, -1)`	1.54	0.90	fails — updates too slowly
Size neutralization	`group_neutralize` by market-cap deciles (`regression_neut` is not available at my level)	1.94	1.25	**self-correlation 0.94 → blocked**





2. The self-correlation wall

BRAIN rejects a new alpha whose correlation with one of my submitted alphas exceeds 0.7, **unless its Sharpe is at least 10% higher**. Every variation above was essentially the same alpha, so better metrics did not help.



3. Finding new ideas

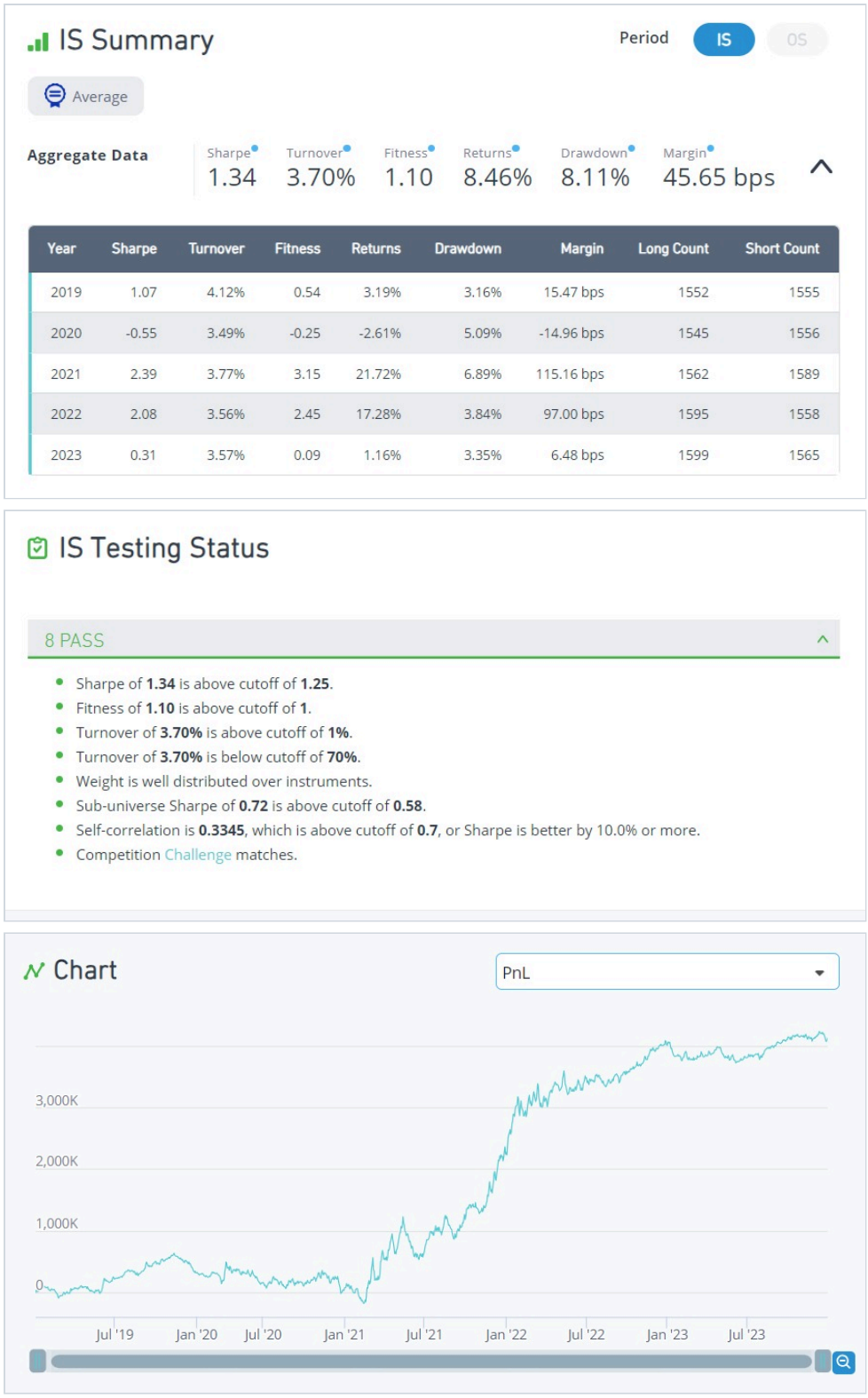
When tuning stopped helping, I searched the web (analyst estimate revisions are an unused data category; a common recipe is signal → scale by volatility → neutralize). I also used BRAIN's built-in **Example** alphas. Note: loading an example also resets the simulation settings.

#	Idea	Sharpe	Outcome
12	20-day change in consensus EPS ÷ price	-0.13	no signal
13	Analyst dispersion (std ÷ mean of estimates)	-0.12	collapsed in 2020
14	`-rank(ebit/capex)` (BRAIN example)	-0.20	fails
15	Funds from operations ÷ long-term debt, 1-year time-series rank	0.67	weak but positive
16→17	Analyst-revision momentum score, then reversed	-0.90 → 0.90	over-reaction to revisions reverts
18	17 + 15, 50:50	1.32	8 checks passed
19	18 with Industry neutralization	**1.34**	**submitted**

Submitted: analyst-revision reversal + financial health ✓

Item	Detail
Idea	Fade stocks whose analyst estimates were revised up most (over-reaction) and buy companies whose cash flow relative to debt is improving versus their own past
Data fields	`composite_factor_score_derivative`, `fnd6_fopo`, `debt_lt`
Settings	TOP3000 · Industry · Decay 8
Result	**Sharpe 1.34 · Fitness 1.10 · Turnover 3.70%** · Margin 45.65 bps
Checks	8 passed. Self-correlation 0.33

> For submitted alphas I share the idea, operators and data fields, but not the exact expression.



Submitted: leverage, revisited ✓

The week 2 alpha `rank(-equity/assets)` (more leveraged companies within a subindustry outperform) was never submitted. On the full 2019–2023 period it scored **Sharpe 1.55, Fitness 1.28, Turnover 1.7%** and passed all eight checks (self-correlation 0.66). Alphas built earlier but never submitted are worth re-checking.



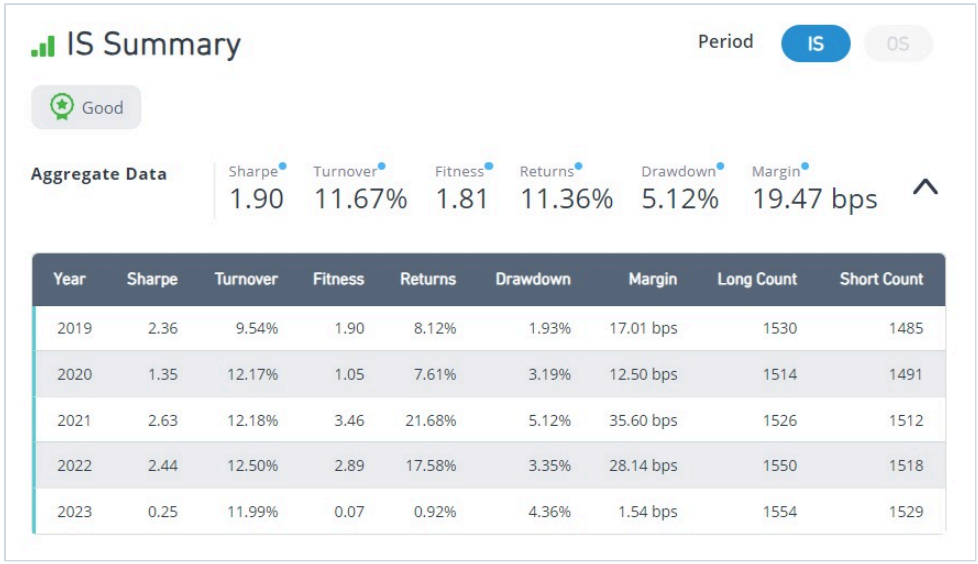
Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.20	1.97%	1.63	6.82%	2.50%	69.36 bps	1417	1531
2020	2.07	1.71%	2.08	12.68%	2.72%	148.05 bps	1407	1539
2021	2.37	1.82%	2.52	14.17%	2.89%	155.93 bps	1393	1525
2022	1.54	1.51%	1.44	10.97%	4.70%	145.32 bps	1446	1526
2023	-0.49	1.38%	-0.20	-2.03%	6.06%	-29.41 bps	1476	1539



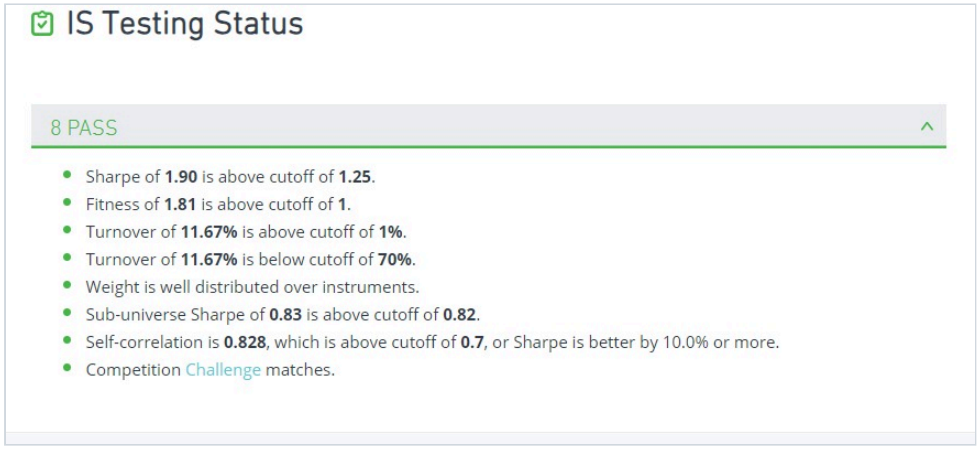
4. Three signals blended → Gold ✓

#	Attempt	Sharpe	Outcome
21	Low-volatility anomaly	0.13	crashed in 2020
22	Increase in fair value of liabilities → short (BRAIN example, 66-day window)	0.91	only ~500 stocks covered
23	22 + option call–put IV gap	1.67	sub-universe 0.47 fails
24	23 tilted toward large caps (× `rank(cap)`)	1.09	fixes sub-universe, kills the signal
25	23 + leverage signal (strong in large caps)	**1.90**	**8 checks passed → submitted**

Item	Detail
Idea	Equal-weight blend of the option traders' bullishness (call–put IV gap), falling liability fair value, and leverage
Chapter 6 concepts	Alpha synergy (combining signals), `rank` for position distribution, NaN-safe `add(…, filter=true)`
Settings	TOP3000 · Subindustry · Decay 8
Result	**Sharpe 1.90 · Fitness 1.81 · Turnover 11.67%** · Returns 11.36% · Drawdown 5.12%
Checks	8 passed. Self-correlation 0.83 passes because its Sharpe is >10% above the correlated alpha; sub-universe 0.83 vs a 0.82 cutoff



Year	Sharpe	Turnover	Fitness	Returns	Drawdown	Margin	Long Count	Short Count
2019	2.36	9.54%	1.90	8.12%	1.93%	17.01 bps	1530	1485
2020	1.35	12.17%	1.05	7.61%	3.19%	12.50 bps	1514	1491
2021	2.63	12.18%	3.46	21.68%	5.12%	35.60 bps	1526	1512
2022	2.44	12.50%	2.89	17.58%	3.35%	28.14 bps	1550	1518
2023	0.25	11.99%	0.07	0.92%	4.36%	1.54 bps	1554	1529





5. After Gold

Reaching Gold unlocks the Research Consultant application. The Quantitative Finance Starter Level Certificate is issued once two onboarding steps in WorldQuant's Workday portal are completed within 20 days: submitting details for a background check, then signing the consulting agreement. Consultants get performance-based compensation, 100,000+ data fields across 8+ regions, the Python API and other advanced features.

Submitted alphas so far

Date	Week · chapter	Ingredients	Idea	Sharpe	Fitness
10/11	1 · Ch2	price + fundamentals	Short-term reversal + profitability	2.46	1.91
10/17	2 · Ch3	fundamentals + price	Earnings-yield time-series rank (TOP200)	1.64	1.76
10/17	2 · Ch4	options + fundamentals	Call–put IV gap + earnings yield	2.49	1.69
10/20	3 · Ch5	news + price	News-conditioned momentum / reversal	1.91	1.25
10/21	4 · Ch6	**analyst** + fundamentals	Revision reversal + financial health	1.34	1.10
10/24	4 · Ch6	fundamentals	Leverage	1.55	1.28
10/25	4 · Ch6	options + fundamentals	IV gap + liability fair value + leverage	1.90	1.81

What I learned this week

1. **A better alpha is not necessarily a submittable one.** Variations of a submitted idea hit self-correlation 0.94. The exception is a Sharpe at least 10% higher than the correlated alpha.
2. **New ideas start weak; blending makes them work.** Twice this week, signals that each failed on their own (0.67–0.91) passed together (1.34, 1.90).
3. **Fix sub-universe failures with a large-cap-friendly signal, not a size tilt.** Multiplying by market cap fixed the check but destroyed the signal.
4. **Re-check old alphas on the full period.** An unsubmitted week 2 alpha passed all checks.
5. **Pace submissions.** Points arrive the day after submission and are capped at 2,000 per day, so one strong alpha per day is the plan.

Next

- Complete the Research Consultant onboarding to receive the certificate
- Use the unlocked data and regions to try ingredients not yet used (volume patterns, other universes)

All experiments

#	Idea	Neut. / Decay	Sharpe	Turnover	Fitness	Note
5	News alpha in TOP1000	Sub / 8	1.44	17%	0.83	
6	News alpha, Industry	Ind / 8	1.73	15%	1.17	
7	News alpha with `signed_power` reversal	Sub / 8	2.03	14%	1.35	
8	#7 with `trade_when`	Sub / 8	1.54	11%	0.90	
9	#7, market-cap group neutralized	Sub / 8	1.94	14%	1.25	self-corr 0.94
10	Buzz regression, 20-day window	Sub / 8	0.58	42%	0.21	
11	Implied ÷ historical volatility, reversed	Sub / 8	-0.79	4%	-0.42	
12	EPS estimate change ÷ price	Sub / 8	-0.13	12%	-0.03	
13	Analyst dispersion	Sub / 8	-0.12	5%	-0.03	
14	EBIT ÷ capex (example)	default	-0.20	1%	-0.09	settings reset
15	FFO ÷ LT debt, time-series rank	Sub / 8	0.67	4%	0.23	
16	Analyst-revision score	Sub / 8	-0.90	2%	-0.63	
17	#16 reversed	Sub / 8	0.90	2%	0.63	
18	#17 + #15	Sub / 8	1.32	4%	1.01	8 passed
19	#18, Industry	Ind / 8	1.34	4%	1.10	submitted
20	Leverage (week 2 alpha)	Sub / 4	1.55	2%	1.28	submitted
21	Low volatility	Sub / 4	0.13	2%	0.06	
22	Liability fair value, reversed	Sub / 4	0.91	7%	0.74	
23	#22 + IV gap	Sub / 8	1.67	18%	1.06	sub-universe fails
24	#23 × rank(cap)	Sub / 8	1.09	14%	0.62	
25	#23 + leverage	Sub / 8	1.90	12%	1.81	submitted → Gold